2014-2015 – Master 2 – Macro I

Lecture 12 : Business Cycle Facts

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Disclaimer

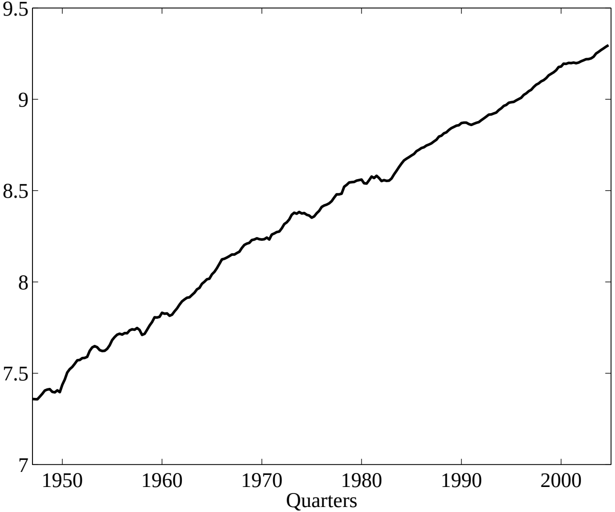
These are the slides I am using in class. They are not self-contained, do not always constitute original material and do contain some "cut and paste" pieces from various sources that I am not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signaling me typos or mistakes at franck.portier@TSE-fr.eu.

Introduction

- Macroeconomics is about the determination of aggregate variables, as measured by national accounts (output, consumption, employment, inflation,...)
- Economists makes a distinction (at least at first pass) between the long run and the short run, between Growth and Business Cycle
- For the methodological part of that lecture, I will consider the U.S.A. as an example.

Introduction

FIGURE 1: US log Real GDP per capita



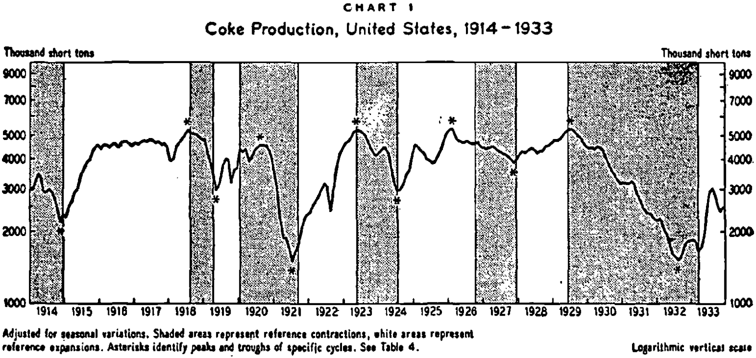
Introduction

- Burns and Mitchell *"Measuring Business Cycles"* (1946, National Bureau of Economic Research) :

    > *"Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises : a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle."*

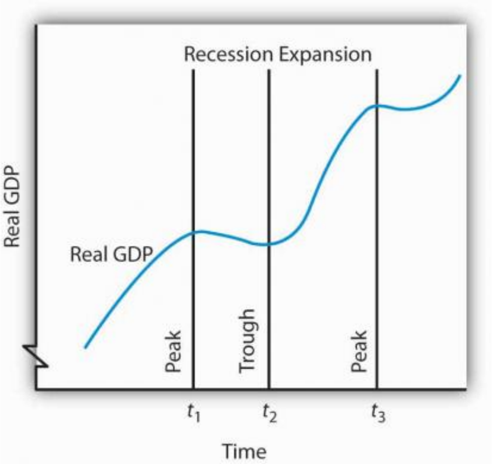
- To identify cycles, B&M assume that they are no shorter than 6 quarters, and found a maximum length of 32 quarters.

FIGURE 2: Reproduced from Burns and Mitchell *Measuring Business Cycles* (1946)



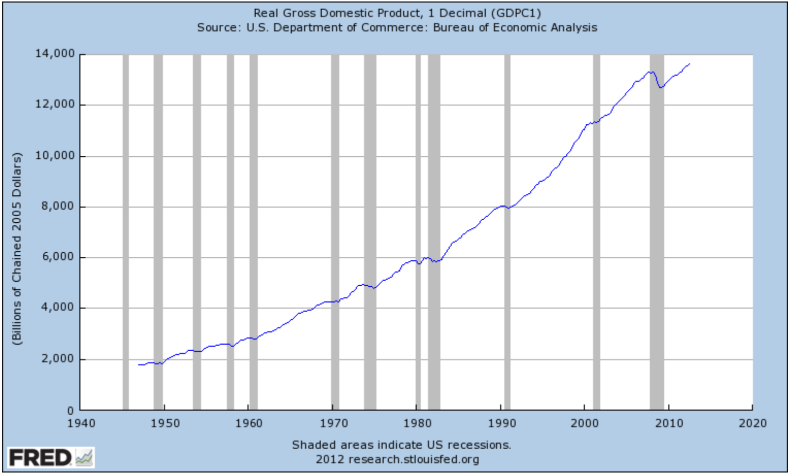
Introduction

FIGURE 3: A Business Cycle



Introduction

FIGURE 4: U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee



Introduction

TABLE 1: Recent U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee

BUSINESS CYCLE REFERENCE DATES		**DURATION IN MONTHS**			
Peak	**Trough**	**Contraction**	**Expansion**	**Cycle**	
Quarterly dates are in parentheses		*Peak to Trough*	*Previous trough to this peak*	*Trough from Previous Trough*	*Peak from Previous Peak*
April 1960(II)	February 1961 (I)	10	24	34	32
December 1969(IV)	November 1970 (IV)	11	106	117	116
November 1973(IV)	March 1975 (I)	16	36	52	47
January 1980(I)	July 1980 (III)	6	58	64	74
July 1981(III)	November 1982 (IV)	16	12	28	18
July 1990(III)	March 1991(I)	8	92	100	108
March 2001(I)	November 2001 (IV)	8	120	128	128
December 2007 (IV)	June 2009 (II)	18	73	91	81

TABLE 2: Average Length of Expansions and Recessions for the U.S. Business Cycles (from the NBER) (in month)

	Contraction	Expansion	Cycle	
	P to T	T to P	T to T	P to P
1854-2009 (33 cycles)	16	42	56	55
1854-1919 (16 cycles)	22	27	48	49
1919-1945 (6 cycles)	18	35	53	53
1945-2009 (11 cycles)	11	59	73	66

- How does the NBER establish this chronology ?
- Here is the "Statement of the NBER Business Cycle Dating Committee on the Determination of the Dates of Turning Points in the U.S. Economy".

  > *"The NBER's Business Cycle Dating Committee maintains a chronology of the U.S. business cycle. The chronology comprises alternating dates of peaks and troughs in economic activity. A recession is a period between a peak and a trough, and an expansion is a period between a trough and a peak. During a recession, a significant decline in economic activity spreads across the economy and can last from a few months to more than a year. Similarly, during an expansion, economic activity rises substantially, spreads across the economy, and usually lasts for several years."*

"In both recessions and expansions, brief reversals in economic activity may occur – a recession may include a short period of expansion followed by further decline ; an expansion may include a short period of contraction followed by further growth. The Committee applies its judgment based on the above definitions of recessions and expansions and has no fixed rule to determine whether a contraction is only a short interruption of an expansion, or an expansion is only a short interruption of a contraction . The most recent example of such a judgment that was less than obvious was in 1980-1982, when the Committee determined that the contraction that began in 1981 was not a continuation of the one that began in 1980, but rather a separate full recession."

Introduction

- Is there a way to translate this into some statistical procedure ?
- What are the data that we shall use and how are they constructed ?
- What are the empirical regularities of BC ?
- These are the questions we will answer in lecture 12.

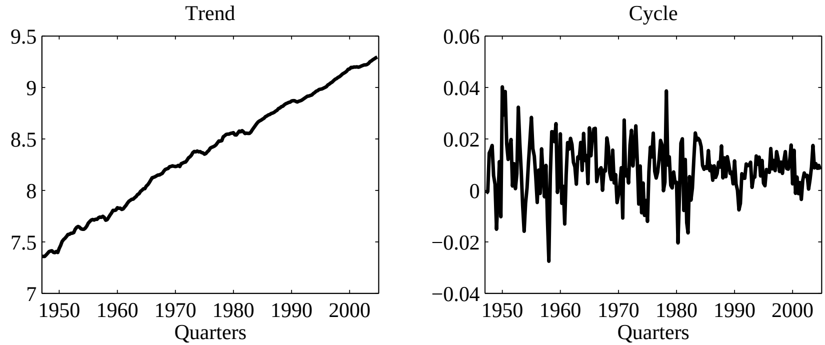
- Any time series $y_t = \log Y_t$ can be decomposed such that

$$y_t = y_t^T + y_t^C$$

- Problem : How to define/identify each component ?
- Several ways of approaching the problem
- Actually : Infinite number of decomposition of a non-stationary process into a cycle and a trend
- Let us see some "intuitive" definition of those decompositions

- Take the growth rate of the series
- Expansion : Positive rate of growth
- Note : the cycle is very volatile (almost iid) – a lot of medium run fluctuations are eliminated

FIGURE 5: US Growth Cycles

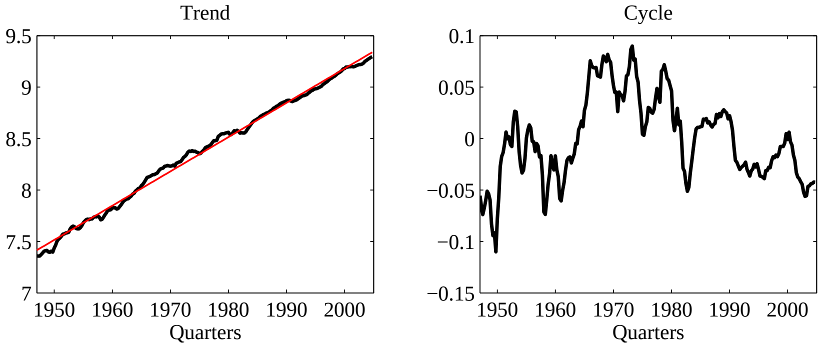


- Deviation from linear trend
- The trend is obtained from linear regression

$$y_t = \alpha + \beta t + u_t$$

- Cycle : $y_t^C = y_t - (\widehat{\alpha} + \widehat{\beta} t)$
- Expansion : Output above the trend

FIGURE 6: US Trend Cycles



- Note : the cycle can be large and very persistent - a lot of medium and long run fluctuations are not eliminated

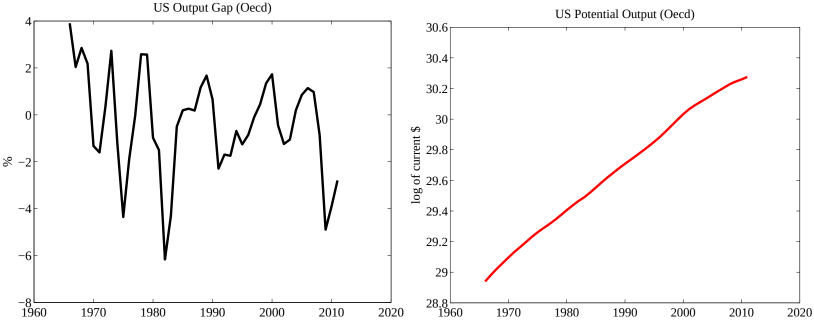
- Define the Output gap as

$$\text{Actual output} - \text{Potential Output}$$

- Expansion : Actual output $\geqslant$ Potential output
- Actual output : easy to observe
- Note : How to identify potential output ? (full utilization ?, efficient ?)
- Example :
  1. estimate $y_t = \alpha \times u_t + \text{other controls} + \varepsilon_t$,
  2. define potential output as $y_t^P = \widehat{\alpha} \times 0\% + \text{other controls} + \widehat{\varepsilon}_t$. (One might choose $u = u^n$ where u^n is the *natural* rate of unemployment (the Oecd chooses the NAIRU (Non Accelerating Inflation Rate of Unemployment))
- Cycle is then $y_t - y_t^P$

- This is an over simplified description of the method used by Oecd.

FIGURE 7: US Output Gap and Potential Output



1. A First Look at Some Methods To Extract the Cycle

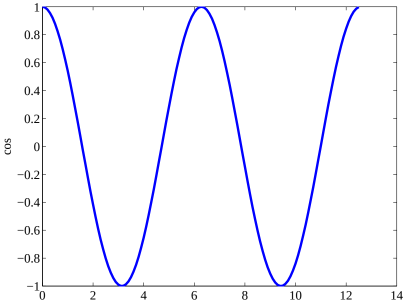
- One could describe many other methods to extract the Business Cycle.
- Ideally, we want to get rid of very short run and long run movements of economic activity.
- The best way to understand this is to decompose economic time series into the frequency domain and filter them.
- For this, we need to understand how a time series can be represented in the frequency domain.
- Here I am giving the main intuitions, a more rigorous treatment will be done in an econometrics course.

- Idea : A series can be seen as the sum of periodic functions.
- A typical periodic function is $\cos(\omega t)$, with period (the time it takes to reproduce itself) $2\pi/\omega$.
  - × Knowing that period of $\cos(t)$ is 2π, for a given t_1, what is the t_2 such that $\cos(\omega t_2) = \cos(\omega t_1)$?
  - × The solution is $t_2 - t_1 = 2\pi/\omega$.
- $\frac{\omega}{2\pi}$ is the *frequency* of oscillation (number of cycles per unit of time)

2. Decomposing a time series into frequency domain

Typical periodic functions

FIGURE 8: Cosine wave with ω=1

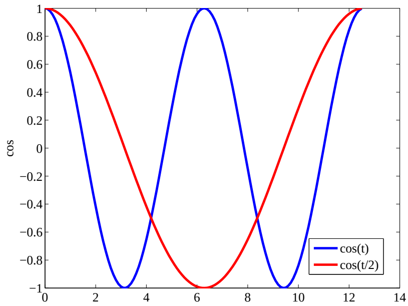


- With $\omega = 1$, the period is $2\pi = 6.28$ and frequency is $\frac{1}{2\pi} = 0.16$.

2. Decomposing a time series into frequency domain

Typical periodic functions

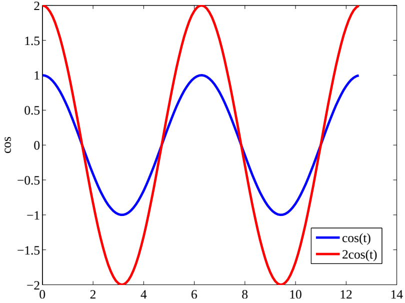
FIGURE 9: Cosine waves with ω=1 or 1/2



- With $\omega = 1/2$, the period is $4\pi = 12.56$ and frequency is $\frac{1}{4\pi} = 0.08$.

2. Decomposing a time series into frequency domain
Typical periodic functions

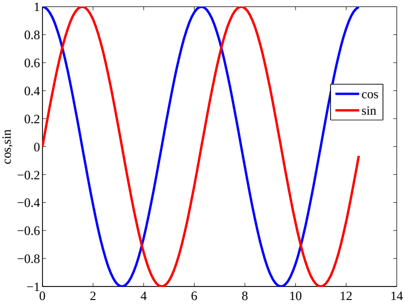
FIGURE 10: Cosine waves with ω=1 and different amplitudes



- Here are plotted $A\cos(t)$ with $A = 1$ or $A = 2$.

- $\sin(\omega t)$ behaves the same way, with same amplitude and period, but with a phase shift

FIGURE 11: Cosine and Sine waves with ω=1



- The idea of spectral decomposition is that with sin and cos, we can span the whole space of covariance stationary time series : the typical periodic function is

$$a\cos(\omega t) + b\sin(\omega t) \tag{1}$$

  whose period is $2\pi/\omega$ but whose phase and amplitude depend on (a, b)
- There is always a sum of type (1) periodic functions that reproduces a given time series
- The spectral density or *spectrum* of a series indicates the weight of each frequency (from low to high) in the total variance of the series.

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- Assume that we observe y_t over T (even) periods, and that it is centered.
- Our goal is to decompose y_t into $T/2$ periodic functions of frequencies $\omega_1, \omega_2, ..., \omega_{T/2}$, with

$$\omega_j = \frac{2\pi j}{T}, \quad j = 1, ..., T/2$$

- Then, we want to write y_t as

$$\begin{aligned} y_t &= a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) \\ &+ a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t) \\ &+ \cdots \\ &+ a_{T/2} \cos(\omega_{T/2} t) + b_{T/2} \sin(\omega_{T/2} t) \end{aligned} \tag{2}$$

for t=1,...,T.

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- Find the T parameters (a_i, b_i)in (2) by OLS.

$$X = \begin{bmatrix} \cos(\omega_1) & sin(\omega_1) & \cdots\cdots & \cos(\omega_{T/2}) & sin(\omega_{T/2}) \\ \cos(2\omega_1) & sin(2\omega_1) & \cdots\cdots & \cos(2\omega_{T/2}) & sin(2\omega_{T/2}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \cos(T\omega_1) & sin(T\omega_1) & \cdots\cdots & \cos(T\omega_{T/2}) & sin(T\omega_{T/2}) \end{bmatrix}$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ \vdots \\ y_{T-1} \\ y_T \end{bmatrix} \qquad \beta = \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ \vdots \\ a_{T/2} \\ b_{T/2} \end{bmatrix}$$

- If we assume $Y = X\beta + u$, we can compute the a and b.

- The a and b coefficients are computed as

$$\beta = (X'X)^{-1}X'Y$$

- Given that we have T explanatory variables for T observations, the R^2 is one and $u = 0$. Here we are just solving a *representation* problem, not an estimation one.
- Note that the last column of X is a column of 0. It is replaced by a column of 1 to deal with non-centered series.

- The coefficient are given by

$$
\begin{aligned}
a_j &= \frac{2}{T}\sum_{t=1}^{T}\cos(\omega_j t)y_t && (3)\\
b_j &= \frac{2}{T}\sum_{t=1}^{T}\sin(\omega_j t)y_t && (4)
\end{aligned}
$$

for $j \leq T/2-1$, and

$$
\begin{aligned}
a_{T/2} &= \frac{1}{T}\sum_{t=1}^{T}\cos(\omega_{T/2} t)y_t && (5)\\
b_{T/2} &= \frac{1}{T}\sum_{t=1}^{T}y_t && (6)
\end{aligned}
$$

- This is of course an approximation of the series.
- The exact formula is :

$$y_t = \int_0^{\pi} (a(\omega)\cos(\omega t) + b(\omega)\sin(\omega t))d\omega \qquad (7)$$

- Any covariance stationary times series process can be represented in the form of (7).

2. Decomposing a time series into frequency domain

Extracting the business cycle (BC) component using the frequency domain representation

- A representation like (7) allow us to make precise the notion of extracting the business cycle component of y_t.
- Assume y_t is observed on a quarterly basis, and that the BC is defined as fluctuations of periods between 6 and 32 quarters (1.5 to 8 years), i.e. for $\omega \in [\underline{\omega} \;\; \overline{\omega}] = [\frac{2\pi}{32}, \frac{2\pi}{6}]$.
- The the BC component of y, denoted y^C, is

$$y_t^C = \int_{\underline{\omega}}^{\overline{\omega}} (a(\omega)\cos(\omega t) + b(\omega)\sin(\omega t))d\omega \qquad (8)$$

- It can be shown that this spectral representation has a time-series equivalent, which is an infinite two-sided moving average of y_t :

$$y_t^C = B_0 + B_1(y_{t-1} + y_{t+1}) + B_2(y_{t-2} + y_{t+2}) + \cdots \qquad (9)$$

with $B_0 = \frac{\overline{\omega} - \underline{\omega}}{\pi}$ and $B_j = \frac{\sin(\overline{\omega} j) - \sin(\underline{\omega} j)}{\pi j}$

$$y_t^C = \int_{\underline{\omega}}^{\overline{\omega}} (a(\omega)\cos(\omega t) + b(\omega)\sin(\omega t))d\omega \tag{8}$$

$$y_t^C = B_0 + B_1(y_{t-1} + y_{t+1}) + B_2(y_{t-2} + y_{t+2}) + \cdots \tag{9}$$

- Why things are not as simple as they look ?
- We have to make an approximation of (9) because it requires an infinity of observations.
- Therefore, the MA is truncated according to some distance criterium

- We consider here *univariate stationary processes*.

Definition 1

The *autocovariance* of a series Y_t is defined as $\lambda_\tau = \text{cov}(Y_t, Y_{t-\tau}) = E(Y_t Y_{t-\tau})$ with the assumption $E(Y_t) = 0\,$;.

Definition 2

For a sequence $a_0, a_1, ..., a_j, ...$, the **generating function** of this sequence is $a(z) = \sum_j a_j z^j$.

- Note : z needs not to have any interpretation
- The generating function (or z-transform) of a process Y_t is $Y(z) = \sum_t Y_t z^t$.

Definition 3

Given a sequence of autocovariance λ_τ, the **autocovariance generating function** is $\lambda(z) = \sum_\tau \lambda_\tau z^\tau$.

- Why is this notation useful ? Consider a stationary process $Y(t)$ with $E(Y_t) = 0$, then define $Y_n(z) = \sum_{t=1}^n Y_t z^t$, then

  $$Y_n(z) Y_n(z^{-1}) = \sum_{t,s} Y_t Y_s z^{t-s}$$

  and

  $$E[Y_n(z) Y_n(z^{-1})] = \sum_{\tau=-1}^{n} (n - |\tau|) \lambda_\tau z^\tau$$

  and therefore

  $$\lambda(z) = \lim_{n \to \infty} \frac{1}{n} E[Y_n(z) Y_n(z^{-1})]s$$

- Correspondence between spectrum and auto-covariogram :
- Assume $z = e^{-i\omega} = \cos\omega - i\sin\omega$ and define

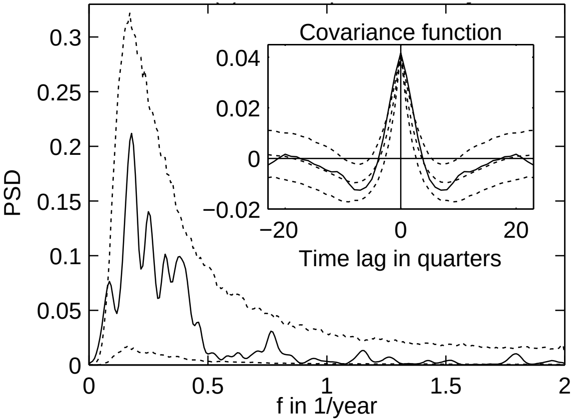
$$s(\omega) = \frac{1}{2\pi}\lambda(z) = \frac{1}{2\pi}\sum_{\tau=-\infty}^{+\infty}\lambda_\tau e^{-i\omega\tau}$$

- Then one can show that

$$\lambda_\tau = \int_{-\pi}^{\pi} e^{i\tau\omega}s(\omega)d\omega \quad \text{and} \quad \lambda_0 = \int_{-\pi}^{\pi} s(\omega)d\omega$$

- The sequence of $\{\lambda_\tau\}$ and $\{s(\omega)\}$ bring the same information.
- The function $\frac{s(\omega)}{\lambda_0}$ has the property of a probability function over $-\pi \leq \omega \leq \pi$: $s(\omega) \geq 0$ and $\int_{-\pi}^{\pi}\frac{s(\omega)}{\lambda_0}d\omega = 1$.
- $s(\omega)$ (rescaled) is the spectral density.
- Next is an estimate of the spectral density, from Groth, Ghil, Hallegatte and Dumas, "Evidence from Genuine Periodicity and Deterministic Causes of US Business Cycles", 2010.

FIGURE 12: A Estimate of US GDP Spectral Density (1954-2005, annual)



Definition 4

Let $\widehat{Y}_t = \sum_{j=0}^{m} c_j Y_{t-j} = C(L) Y_t$. $\widehat{Y}_t$ is a **filtered** version of Y_t.

- One can show that the spectral density of $\widehat{Y}_t$ is

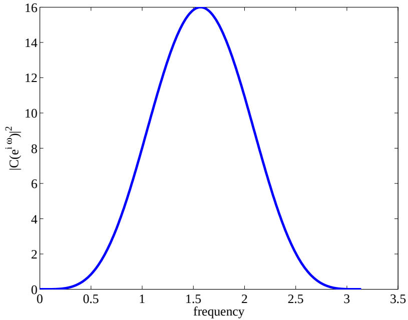
$$s_{\widehat{y}}(\omega) = C(e^{-i\omega})C(e^{i\omega})s_y(\omega) = |C(e^{i\omega})|^2 s_y(\omega)$$

.

- $c_0 = 1, c_2 = -2, c_4 = -1$ and $c_j = 0$ for other j.
- $\widehat{Y}_t = C(L)Y_t = Y_t - 2(Y_{t-2} + Y_{t+2}) - (Y_{t-4} + Y_{t+4})$
- Then

$$|C(e^{i\omega})|^2 = 4(1 - \cos 2\omega)^2$$

FIGURE 13: A Band Pass Filter



- Here is an example of the use of Band Pass filters from Roberto Pancrazi, "Spectral Covariance Instability Test : An Application to the Great Moderation", TSE 2010.
- High Frequency : periodicity between 2 and 32 quarters
- Medium Frequency : periodicity between 32 and 80 quarters
- High to Medium Frequency : periodicity between 2 and 80 quarters

FIGURE 14: US GDP

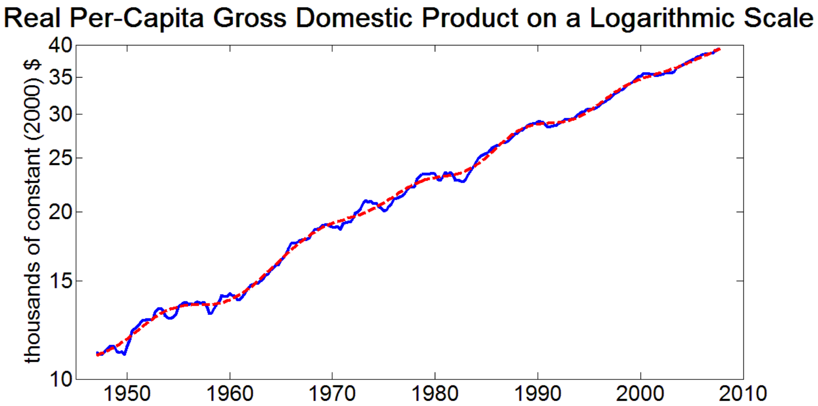
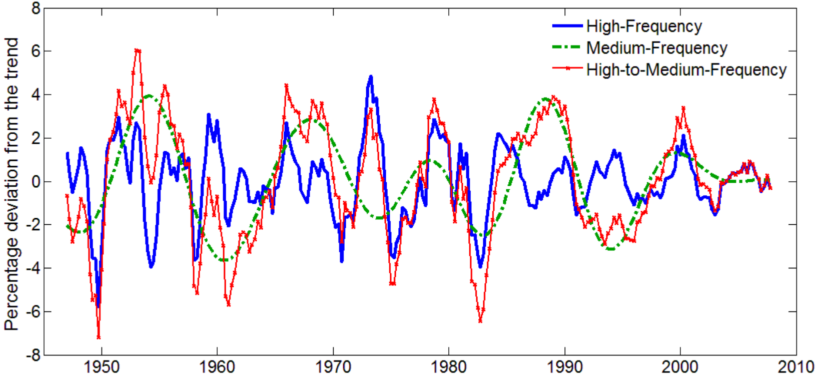


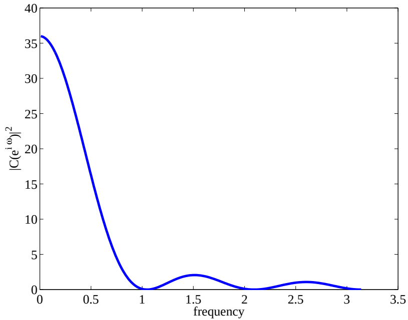
FIGURE 15: Various Cycles



- $\widetilde{Y}_t = \sum_{j=0}^{m} Y_{t-j}$. Then

$$|C(e^{i\omega})|^2 = \frac{1 - \cos(m+1)\omega}{1 - \cos\omega}$$

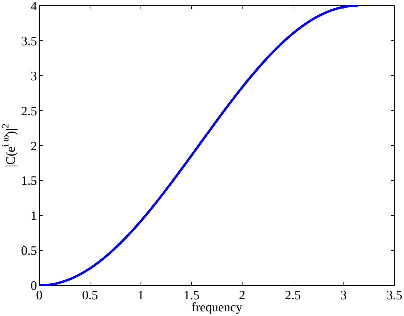
FIGURE 16: A Low Pass Filter



- First difference $\widetilde{Y}_t = (1 - L)Y_t$. Then

$$|C(e^{i\omega})|^2 = 2 - 2\cos\omega$$

FIGURE 17: First Difference



- Very popular in the macro literature
- In the time domain, the idea is to remove a trend which is smooth, but not linear
- The trend y_t^T is the Argmin of :

$$\sum_{t=1}^{T}(y_t - y_t^T)^2 + \lambda \sum_{t=2}^{T}((y_{t+1}^T - y_t^T) - (x_t - x_{t-1}))^2$$

- if $\lambda = +\infty$, it is linear detrending.
- The solution of this program solves $y_t/\lambda = y_{t+2}^T - 4y_{t+1}^T + (6+\lambda)y_t^T - 4y_{t-1}^T - y_{t-2}^T$
- The solution is a symmetric MA of order $+\infty$:

$$y_t^T = \sum_{j=-\infty}^{\infty} a_{|j|} y_{t+j}$$

- Then $y_t^C = y_t - y_t^T$ is a time invariant linear symmetric filter.

- With $\lambda = 1600$ on quarterly data, it removes cycles of period greater than 10 years.
- The transfer function is

$$|C(e^{i\omega})|^2 = \frac{16\lambda^2(1-\cos\omega)^4}{(1+4\lambda(1-\cos\omega)^2)^2}$$

FIGURE 18: HODDRICK-PRESCOTT Filter

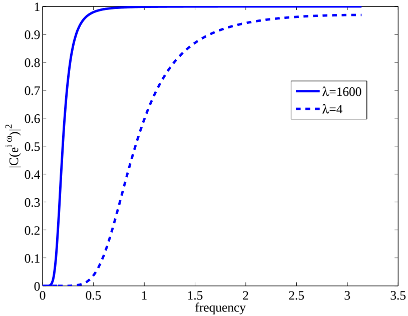
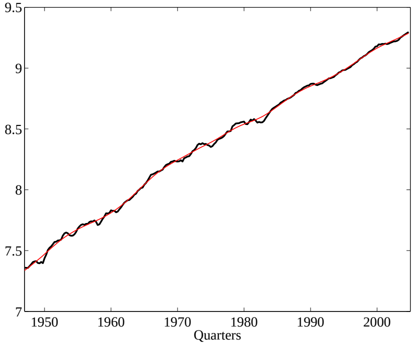


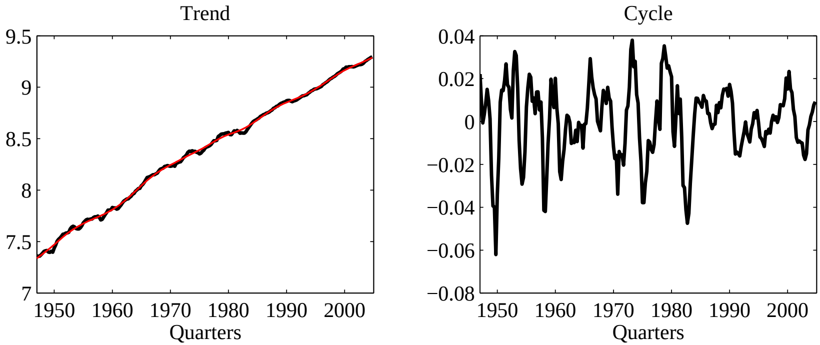
FIGURE 19: US HP Trend



2. Decomposing a time series into frequency domain

The HP filter at work

Figure 20: US HP Cycle



3. Quick Overview of National Accounts

- Data : we mainly consider aggregate quantities of goods and services and prices, labor market statistics and interest rates.
- Aggregate quantities of goods and services and prices mostly come from national accounts.
- Decent level of harmonization across countries (System of National Accounts (SNA) promoted by the United Nations)
- from the UN Handbook of National Accounting :

  > *"The System of National Accounts (SNA) helps economists to measure the level of economic development and the rate of economic growth, the change in consumption, saving, investment, debts and wealth (or net worth) for not only the total economy but also each of its institutional sectors (such as government, public and private corporations, households and non-profit institutions serving households)"*

- I present here the basic concepts – mainly about definitions and conventions ("accounting")

- For an economy, the total supply of goods and services must equal the total uses
  `total supply of goods and services = total uses of goods and services`
- Expanding each side :
  `output + imports = intermediate consumption + final consumption + gross capital formation + exports`
- Note 1 : Intermediate consumption consists of the goods and services consumed in the production process (excluding the consumption of fixed assets)
- Note 2 : Final consumption consists of the goods and services provided to the benefit of final consumers.

- Gross value added (leave for a moment the issue of taxes and subsidies on goods and services aside)

  ```
  gross value added = output - intermediate consumption
  = final consumption + gross capital formation +
  exports - imports
  ```
- Final consumption and gross fixed capital formation are measured from the perspective of the consumer or purchaser. Their values take into account the taxes and subsidies on goods and services.
- Output is measured from the perspective of producers in terms of the receipts receivable by them, leaving all of the taxes on goods and services aside while including subsidies on goods and services.

- Therefore, taxes on goods and services have to be added to output and subsidies subtracted from output in order to arrive at a uniform valuation of supply and uses.

```
output + taxes - subsidies - intermediate consumption
= final consumption + gross capital formation +
exports - imports
```

- GDP can be measured by having the values for output and intermediate consumption aggregated across the various industries of an economy. This is GDP by production approach :

  ```
  GDP = output + taxes - subsidies - intermediate
  consumption = gross value added + taxes - subsidies
  ```

- Gross domestic product can also be viewed as the value of all goods and services available for different domestic final uses or for exports. This is GDP by expenditure approach :

  ```
  GDP = final consumption + gross capital formation +
  exports - imports
  ```

- The production process creates incomes for not only the owners of the inputs used in production but also for owners of capital and for the government. The value of those incomes is equal to gross domestic product. Hence, GDP can also be calculated as the sum of compensation of employees, taxes less subsidies and gross operating surplus/mixed income. This is the GDP by income approach :

  ```
  GDP = compensation of employees + taxes - subsidies +
  gross operating surplus / mixed income
  ```

3. Quick Overview of National Accounts

Gross national income

- Gross domestic product refers to production of all resident units within the borders of a country, which is not exactly the same as the production of all productive activities of residents :
  GNI = GDP + compensation of employees and property income from the rest of the world
  - compensation of employees and property income to the rest of the world
- All GNI is not available for final uses domestically since some of it is transferred to other countries without anything being received in exchange (for example remittances)
  gross national disposable income = GNI + current transfers from the rest of the world - current transfers to the rest of the world
- Gross national disposable income is the income available for consumption and saving :

- Gross saving together with net capital transfers (capital transfers receivable less capital transfers payable) from the rest of the world provides the resources for investment in non-financial assets, which is called gross capital formation.
- Gross capital formation = the net acquisition of fixed assets, such as residential and non-residential buildings, plants and equipments, the net acquisition of valuables and/or the increase in inventories.
- The difference between gross saving plus net capital transfers and gross capital formation is net borrowing or net lending from the rest of the world, depending whether uses exceed resources or vice versa.

  `gross saving = gross national disposable income - final consumption`

  and

  `net lending (+) / net borrowing (-) = gross saving + net capital transfers - gross capital formation`

- Net borrowing / net lending is also reflected in transactions in financial assets and liabilities with the rest of the world. It is equal to the difference between net acquisition of financial assets and net incurrence of liabilities (foreign exchange, bonds, loans etc.) :
  `net lending (+) / net borrowing (-) = net acquisition of financial assets - net incurrence of liabilities`

- Net worth is the difference between the total value of non-financial and financial assets and the total value of liabilities of an economy. It is a measure of the net wealth of a nation. Change in net worth measures the change in wealth of a nation.
- Net capital transfers from abroad are equal to gross capital formation less consumption of fixed capital and plus net lending (+)/net borrowing (-) from the rest of the world.

3. Quick Overview of National Accounts

Summary

TABLE T1.1. SIMPLIFIED SEQUENCE OF ACCOUNTS OF THE DOMESTIC ECONOMY

		Uses	Resources
	Production account		
	Output of goods and services		100
Less	Intermediate consumption	40	
Equals	**Gross value added/GDP**	**60**	
	Primary distribution of income account		
	Gross value added/GDP		60
Plus	Compensation of employees and property income receivable from the rest of the world (ROW)		4
Less	Compensation of employees and property income payable to ROW	1	
Equals	**Gross national income**	**63**	
	Secondary distribution of income account		
	Gross national income		63
Plus	Current transfers receivable from ROW		1
Less	Less current transfers payable to ROW	2	
Equals	**Gross disposable income**	**62**	
	Use of income account		
	Gross disposable income		62
Less	Final consumption	40	
Equals	**Gross saving**	**22**	
		Uses	**Resources**
	Capital account		
	Gross saving		22
Less	Gross capital formation	15	
Plus	Capital transfers from ROW		1
Less	Capital transfers to ROW	1	
Equals	**Net lending to ROW**	**7**	
	Financial account	**Changes in assets**	**Changes in liabilities**
	Net acquisition of financial assets		
	Money	3	
	Loans	4	
Less	Net incurrence of liabilities		0
Equals	**Net lending to ROW**		**7**
	Changes in the balance sheet due to transactions	**Assets**	**Liabilities**
	Non-financial assets		
	Gross capital formation	15	
	Consumption of fixed capital	-1	
Less	Financial assets/financial liabilities	7	0
Equals	**Net worth**		21

Definition 5

Output is the value of the goods and services which are produced by an establishment in the economy that become available for use outside that establishment

Definition 6

Intermediate consumption includes goods and services which are entirely used up by producers in the course of production to produce output of goods and services during the accounting period.

Definition 7

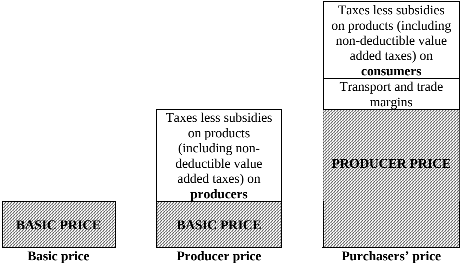
Final consumption includes goods and services which are used by households or the community to satisfy their individual wants and social needs. Consumption is broken down into a) Final consumption expenditure of households ; b) Final consumption expenditure of general government ; c) Final consumption expenditure of non-profit institutions serving households.

- For households, all consumed goods, whether durable (cars, refrigerators, air-conditioners etc.) or non-durable (food, clothes), are part of final consumption, with the exception of purchases for own-construction or improvements of residential housing, which are treated as part of gross capital formation.

Definition 8

Gross capital formation in SNA is the same as the concept of investment in capital goods used by economists. It includes only produced capital goods (machinery, buildings, roads, artistic originals etc.) and improvements to non-produced assets. Gross capital formation measures the additions to the capital stock of buildings, equipment and inventories, i.e., the additions to the capacity to produce more goods and income in the future.

- Outputs, whether or not sold, are valued at market or "equivalent market prices".
- 3 types of market prices of the same good due to taxes and subsidies.
- Basic price is the amount receivable by the producer from the purchasers for a unit of output. Then Producer price and Purchasers price are defined



- Output is recommended to be measured at production costs when products have no market price.

- To compare quantities of two different years, one needs to adjust for changes in prices, to *deflate* nominal (current dollars) measures in order to obtain real (constant dollars) quantities.
- This is done (basically) by choosing a base year (year N). The real quantities of year $N+1$ are then multiplied by their price in year N to compute constant dollar measures (in dollars of year N).
- This is easy for potatoes (always the same good), not so for computers or cars (improvement in quality).

- Lucas' definition :

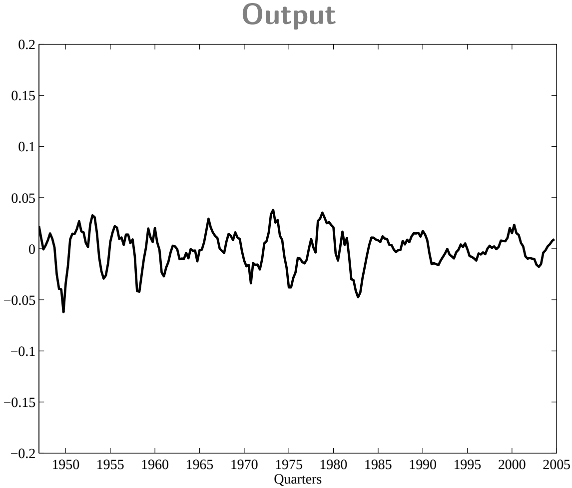
  "Recurrent **fluctuations** *of macroeconomic aggregates around trend"*

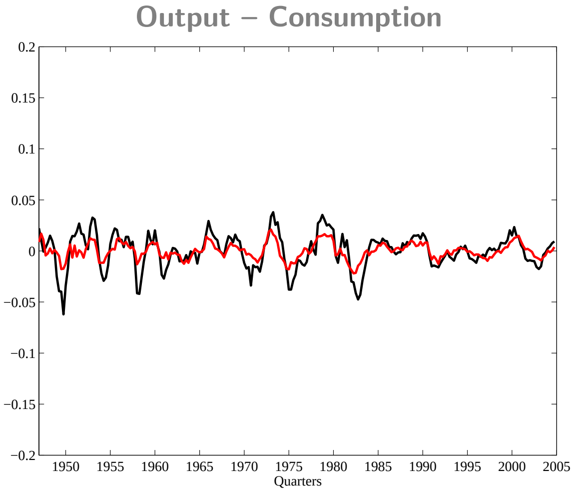
- Want to find **regularities** (*Stylized facts*)

- Business Cycles are characterized by a set of statistics :
  - × Volatilities of time series (standard deviations)
  - × Comovements of time series (correlations, serial correlations)
- Why only looking at the US ?

  *"Though there is absolutely no theoretical reason to anticipate it, one is led by the facts to conclude that, with respect to the qualitative behavior of co-movements among series, business cycles are all alike." (*LUCAS *1977)*

- Consumption (C) : Nondurables + Services
- Investment (I) : Durables + Fixed Investment + Changes in inventories
- Government spending (G)
- Output : $C + I + G$
- Labor : hours worked
- Labor Productivity : Output / Labor

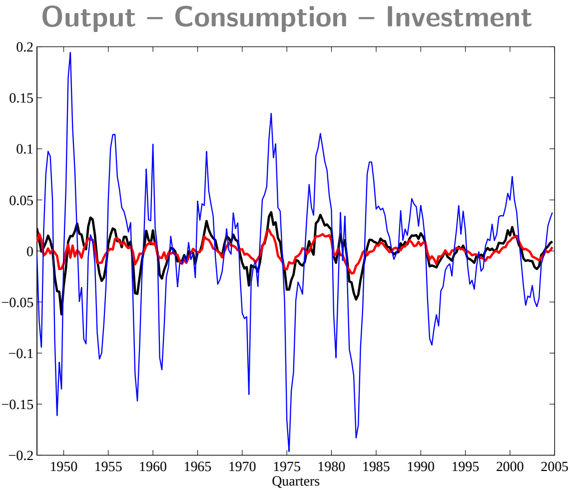

Output
0.2
0.15
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
Quarters

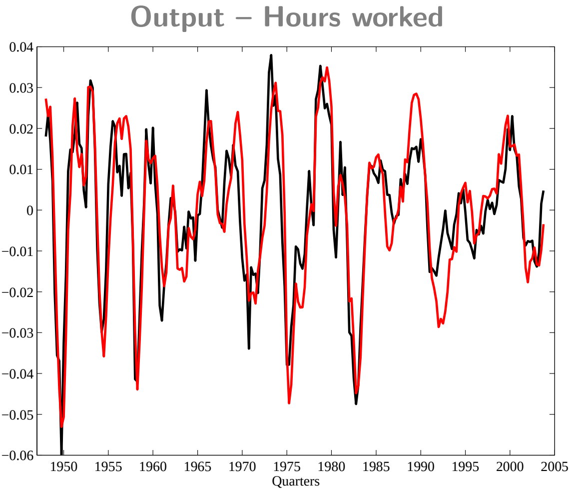


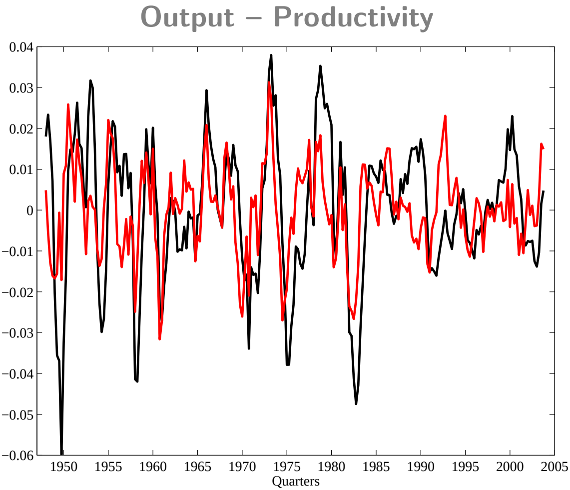
Output – Consumption
0.2
0.15
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
Quarters

4. U.S. Business Cycles

Main Real Aggregates

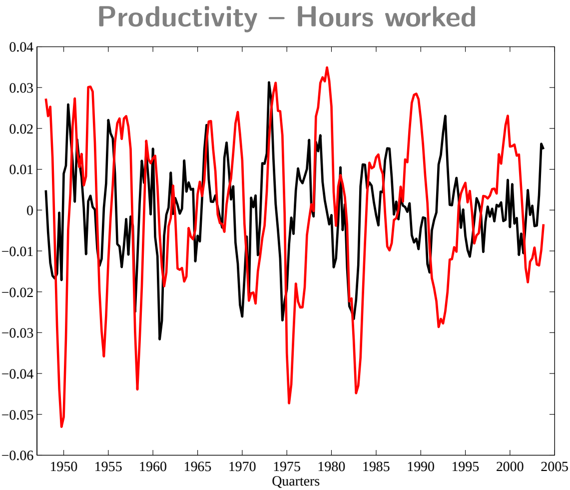



Output – Hours worked
0.04
0.03
0.02
0.01
0
−0.01
−0.02
−0.03
−0.04
−0.05
−0.06
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
Quarters


Output – Productivity
0.04
0.03
0.02
0.01
0
−0.01
−0.02
−0.03
−0.04
−0.05
−0.06
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
Quarters

4. U.S. Business Cycles

Main Real Aggregates



- Those figures are taken from STOCK and WATSON, "Business Cycle Fluctuations in US Macroeconomics Time Series", chapter 1 of the Handbook of Macroeconomics, 1999
- Quarterly US data, 1947-1997

More (Much More) Data

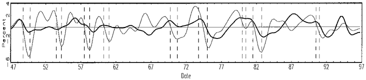


Fig. 3.3. Finance, insurance and real estate employment.

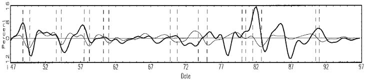


Fig. 3.4. Mining employment.

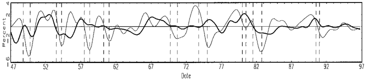


Fig. 3.5. Government employment.

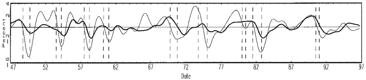


Fig. 3.6. Service employment.

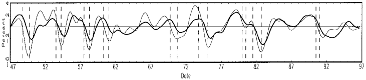


Fig. 3.7. Wholesale and retail trade employment.

More (Much More) Data

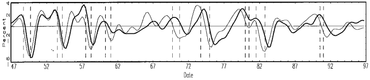


Fig. 3.8. Transportation and public utility employment.

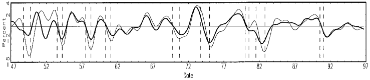


Fig. 3.9. Consumption (total).

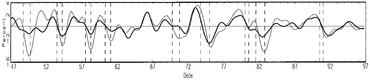


Fig. 3.10. Consumption (nondurables).

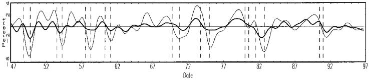


Fig. 3.11. Consumption (services).

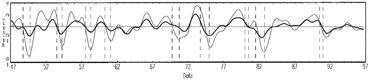


Fig. 3.12. Consumption (nondurables + services).

4. U.S. Business Cycles

More (Much More) Data

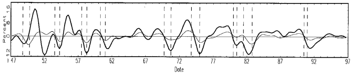


Fig. 3.13. Consumption (durables).

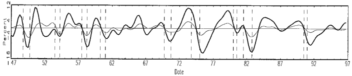


Fig. 3.14. Investment (total fixed).

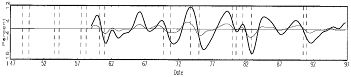


Fig. 3.15. Investment (equipment).

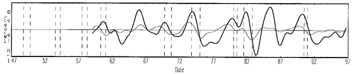


Fig. 3.16. Investment (nonresidential structures).

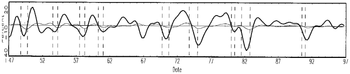


Fig. 3.17. Investment (residential structures).

4. U.S. Business Cycles

More (Much More) Data

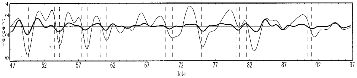


Fig. 3.18. Change in business inventories (relative to trend GDP).

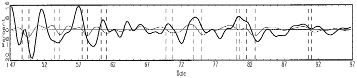


Fig. 3.19. Exports.

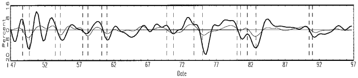


Fig. 3.20. Imports.

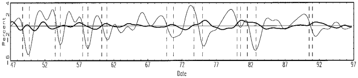


Fig. 3.21. Trade balance (relative to trend GDP).

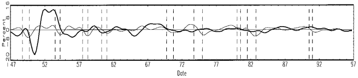


Fig. 3.22. Government purchases.

4. U.S. Business Cycles

More (Much More) Data

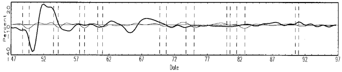


Fig. 3.23. Government purchases (defense).

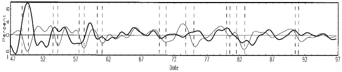


Fig. 3.24. Government purchases (non-defense).

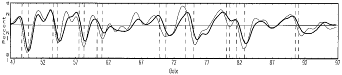


Fig. 3.25. Employment (total employees).

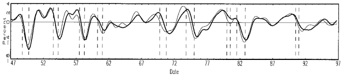


Fig. 3.26. Employment (total hours).

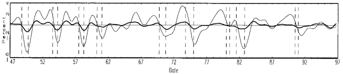


Fig. 3.27. Employment (average weekly hours).

More (Much More) Data

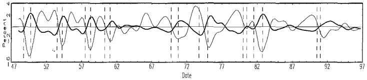


Fig. 3.28. Unemployment rate.

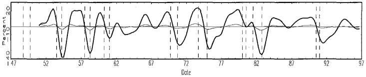


Fig. 3.29. Vacancies (Help Wanted index).

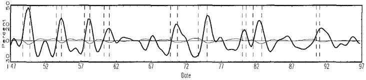


Fig. 3.30. New unemployment claims.

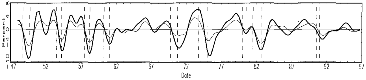


Fig. 3.31. Capacity utilization.

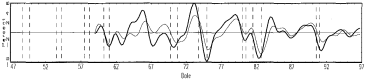


Fig. 3.32. Total factor productivity.

4. U.S. Business Cycles

More (Much More) Data

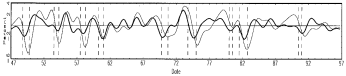


Fig. 3.33. Average labor productivity.

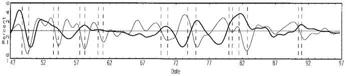


Fig. 3.34. Consumer price index (level).

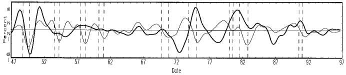


Fig. 3.35. Producer price index (level).

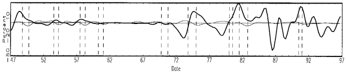


Fig. 3.36. Oil prices.

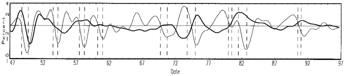


Fig. 3.37. GDP price deflator (level).

4. U.S. Business Cycles

More (Much More) Data

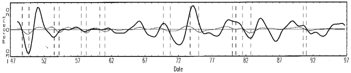


Fig. 3.38. Commodity price index (level).

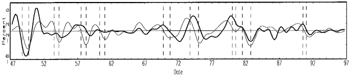


Fig. 3.39. Consumer price index (inflation rate).

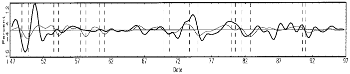


Fig. 3.40. Producer price index (inflation rate).

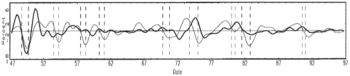


Fig. 3.41. GDP price deflator (inflation rate).

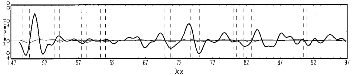


Fig. 3.42. Commodity price index (inflation rate).

More (Much More) Data

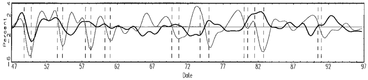


Fig. 3.43. Nominal wage rate (level).

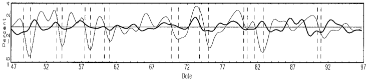


Fig. 3.44. Real wage rate (level).

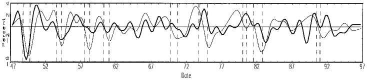


Fig. 3.45. Nominal wage rate (rate of change).

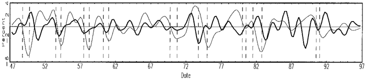


Fig. 3.46. Real wage rate (rate of change).

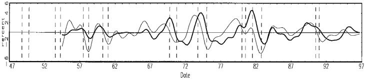


Fig. 3.47. Federal funds rate.

4. U.S. Business Cycles

More (Much More) Data

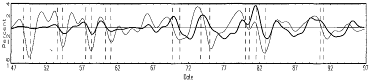


Fig. 3.48. Treasury Bill rate (3 month).

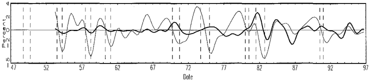


Fig. 3.49. Treasury Bond rate (10 year).

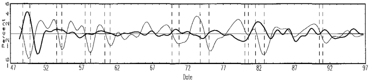


Fig. 3.50. Real Treasury Bill rate (3 month).

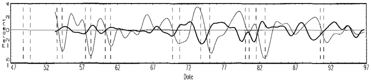


Fig. 3.51. Yield curve spread (long–short).

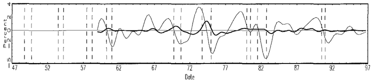


Fig. 3.52. Commercial paper/Treasury Bill spread.

4. U.S. Business Cycles

More (Much More) Data

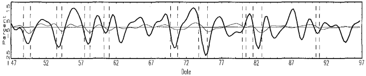


Fig. 3.53. Stock prices.

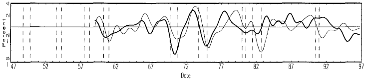


Fig. 3.54. Money stock (M2, nominal level).

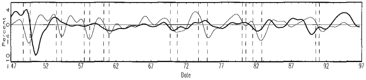


Fig. 3.55. Monetary base (nominal level).

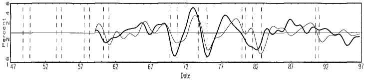


Fig. 3.56. Money stock (M2, real level).

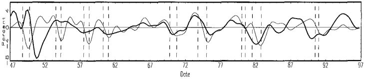


Fig. 3.57. Monetary base (real level).

More (Much More) Data

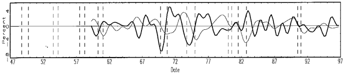


Fig. 3.58. Money stock (M2, nominal rate of change).

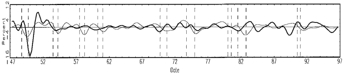


Fig. 3.59. Monetary base (Nominal rate of change).

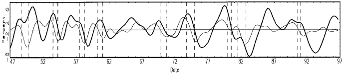


Fig. 3.60. Consumer credit.

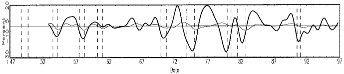


Fig. 3.61. Consumer expectations.

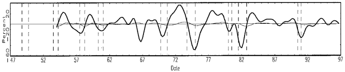


Fig. 3.62. Building permits.

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More (Much More) Data

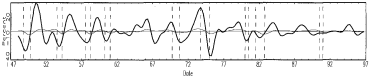

Fig. 3.63. Vendor performance.

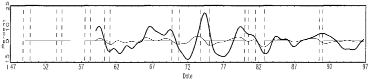

Fig. 3.64. Manufacturers' unfilled orders, durable goods industry.

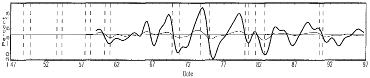

Fig. 3.65. Manufacturers' new orders, non-defense capital goods.

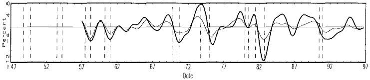

Fig. 3.66. Industrial production, Canada.

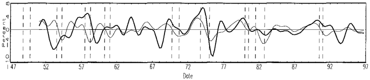

Fig. 3.67. Industrial production, France.

4. U.S. Business Cycles

More (Much More) Data

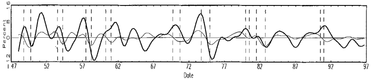

Fig. 3.68. Industrial production, Japan.

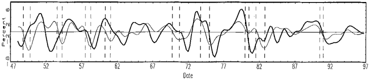

Fig. 3.69. Industrial production, UK.

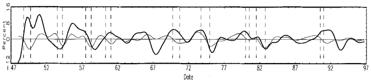

Fig. 3.70. Industrial production, Germany.

- We want to characterize fluctuations $\rightsquigarrow$ amplitude and movements
- Amplitude : volatilities $\rightsquigarrow$ standard deviations
- Comovements : correlations

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	Serial(1)
Output	1.70	–	–	–	0.84
Consumption	0.80	0.47	0.78	–	0.83
Services	1.11	0.66	0.72	–	0.80
Non Durables	0.72	0.42	0.71	–	0.77
Investment	6.49	3.83	0.84	–	0.81
Fixed investment	5.08	3.00	0.80	–	0.88
Durables	5.23	3.09	0.58	–	0.72
Changes in inventories	22.48	13.26	0.48	–	0.40
Hours worked	1.69	1.00	0.86	–	0.89
Labor productivity	0.90	0.53	0.41	0.09	0.69

1. Consumption (of non-durables) is less volatile than output
2. Investment is more volatile than output
3. Hours worked are as volatile as output
4. Capital is much less volatile than output
5. Labor productivity is less volatile than output
6. Real wage is much less volatile than output
7. All those variables are persistent and procyclical except Labor productivity that is acyclical

- Quoting Lucas 1977 "Understanding Business Cycles"
    1. Output movements across broadly defined sectors move together.
    2. Production of producer and consumer durables exhibits much greater amplitude than does the production of nondurables
    3. Production and prices of agricultural goods and natural resources have lower than average conformity.
    4. Business profits show high conformity and much greater amplitude than other series.
    5. Prices generally are procyclical.
    6. Short-term interest rates are procyclical ; long-term rates slightly so.
    7. Monetary aggregates and velocity measures are procyclical.

- From FIORITO and KOLLINTZAS, "Stylized facts of business cycles in the G7 from a real business cycles perspective", European Economic Review, 1994.
- Quarterly data 1960-1989

5. International Business Cycles

Within countries

Cross correlations of real GNP/GDP with the components of spending, income, and output in levels. [a,b]

Variable	Vol.	X_{t-5}	X_{t-4}	X_{t-3}	X_{t-2}	X_{t-1}	X_t	X_{t+1}	X_{t+2}	X_{t+3}	X_{t+4}	X_{t+5}
(1) Real GNP/GDP												
USA	1.74	0.01	0.21	0.41	0.65	0.85	1.0	0.85	0.65	0.41	0.21	0.01
Canada	1.39	−0.12	0.04	0.27	0.51	0.78	1.0	0.78	0.51	0.27	0.04	−0.12
Japan	1.53	0.02	0.19	0.38	0.59	0.78	1.0	0.78	0.59	0.38	0.19	0.02
Germany	1.69	−0.02	0.23	0.35	0.46	0.67	1.0	0.67	0.46	0.35	0.23	−0.02
France	0.90	−0.06	0.10	0.30	0.54	0.77	1.0	0.77	0.54	0.30	0.10	−0.06
UK	1.54	−0.02	0.07	0.20	0.37	0.55	1.0	0.55	0.37	0.20	0.07	−0.02
Italy	1.70	−0.21	−0.04	0.22	0.52	0.80	1.0	0.80	0.52	0.22	−0.04	−0.21
(2) Consumption expenditure												
US	1.29	0.32	0.48	0.59	0.72	0.79	0.80	0.63	0.43	0.22	0.00	−0.17
Canada	1.27	−0.08	0.16	0.40	0.57	0.72	0.79	0.65	0.44	0.27	0.06	−0.03
Japan	1.33	−0.10	0.08	0.28	0.42	0.56	0.72	0.54	0.40	0.22	0.01	−0.11
Germany	1.53	0.11	0.26	0.37	0.46	0.58	0.69	0.55	0.49	0.38	0.32	0.21
France	0.86	−0.27	0.42	−0.63	0.73	0.72	0.62	0.30	0.10	−0.14	0.25	−0.32
UK	1.67	0.03	0.13	0.30	0.39	0.46	0.67	0.42	0.38	0.26	0.10	0.08
Italy	1.19	−0.15	0.07	0.34	0.59	0.74	0.78	0.69	0.50	0.25	0.03	−0.15
(3) Fixed investment												
US	5.57	0.14	0.30	0.47	0.67	0.83	0.90	0.78	0.59	0.35	0.12	−0.09
Canada	4.60	−0.43	−0.29	−0.07	0.18	0.40	0.53	0.52	0.41	0.32	0.21	0.14
Japan	4.57	−0.11	0.04	0.23	0.45	0.64	0.83	0.78	0.69	0.51	0.29	0.05

Cross correlations of real GNP/GDP with the components of spending, income, and output in levels. [a,b]

Variable	Vol.	X_{t-5}	X_{t-4}	X_{t-3}	X_{t-2}	X_{t-1}	X_t	X_{t+1}	X_{t+2}	X_{t+3}	X_{t+4}	X_{t+5}
Germany	4.90	0.04	0.26	0.37	0.42	0.60	0.84	0.54	0.42	0.37	0.29	0.12
France	2.70	−0.11	0.06	0.26	0.46	0.66	0.78	0.69	0.57	0.41	0.25	0.13
UK	3.57	−0.11	−0.04	0.08	0.23	0.33	0.60	0.53	0.38	0.31	0.23	0.05
Italy	4.88	−0.16	−0.00	0.23	0.47	0.70	0.88	0.81	0.67	0.47	0.25	0.05
(5) Equipment investment												
US	6.28	−0.13	0.02	0.21	0.46	0.68	0.86	0.87	0.77	0.59	0.38	0.18
Canada	7.13	−0.49	−0.35	−0.18	0.03	0.27	0.43	0.51	0.53	0.50	0.34	0.25
Japan	5.96	−0.09	0.02	0.17	0.38	0.58	0.74	0.73	0.69	0.54	0.34	0.14
Germany	6.09	0.12	0.36	0.48	0.52	0.61	0.73	0.58	0.49	0.39	0.23	0.09
France	3.90	0.08	−0.23	0.39	0.58	0.70	0.74	0.53	0.31	0.12	−0.06	−0.17
UK	4.88	−0.12	−0.07	0.05	0.21	0.38	0.56	0.51	0.47	0.44	0.32	0.25
Italy	7.92	−0.15	0.01	0.25	0.48	0.69	0.85	0.74	0.57	0.38	0.14	−0.05
(6) Construction investment												
US	6.26	0.31	0.45	0.57	0.70	0.80	0.78	0.58	0.35	0.11	−0.10	−0.27
Canada	3.83	−0.23	−0.12	0.10	0.34	0.50	0.55	0.41	0.18	0.06	0.01	−0.04
Japan	5.58	−0.04	0.09	0.23	0.31	0.32	0.43	0.35	0.18	0.07	−0.05	−0.18
Germany	5.56	0.00	0.15	0.22	0.27	0.47	0.72	0.40	0.28	0.27	0.25	0.10
France	2.49	−0.25	−0.11	0.08	0.25	0.48	0.65	0.65	0.65	0.54	0.45	0.33
UK	3.90	0.15	0.19	0.28	0.26	0.21	0.38	0.27	0.08	−0.00	−0.08	−0.24
Italy	3.57	−0.11	0.00	0.18	0.36	0.57	0.74	0.74	0.65	0.50	0.36	0.20
(7) Inventory changes												
US	18.2	−0.01	0.08	0.22	0.35	0.49	0.64	0.48	0.26	0.03	−0.14	−0.30
Canada	35.4	0.07	0.15	0.25	0.43	0.60	0.68	0.53	0.33	0.06	−0.18	−0.32
Japan	45.4	−0.05	−0.03	0.07	0.23	0.38	0.38	0.38	0.25	0.20	0.20	0.10
Germany	49.2	0.07	0.19	0.31	0.32	0.33	0.35	0.29	0.14	0.02	−0.13	−0.27
France	30.1	−0.15	−0.09	−0.04	0.05	0.22	0.47	0.44	0.25	0.16	−0.05	−0.27
UK	26.6	0.03	0.12	0.16	0.26	0.42	0.55	0.38	0.19	0.00	−0.08	−0.17
Italy	66.8	−0.07	0.10	0.21	0.39	0.51	0.56	0.32	0.00	−0.24	−0.41	−0.48
(8) Government final consumption												
US	1.98	−0.07	−0.04	0.00	0.06	0.11	0.19	0.24	0.27	0.30	0.35	0.37
Canada	1.46	−0.18	−0.20	−0.24	−0.23	−0.20	−0.12	−0.09	−0.08	0.05	0.14	0.18
Japan	2.89	0.25	0.33	0.30	0.28	0.30	0.32	0.04	−0.05	−0.08	−0.05	−0.06
Germany	1.47	−0.19	−0.11	−0.13	−0.10	−0.06	0.05	0.06	0.16	0.23	0.36	0.41
France	0.70	0.46	0.61	0.56	0.46	0.32	0.18	−0.07	−0.23	−0.31	−0.30	−0.24
UK	1.43	−0.09	−0.03	−0.07	−0.06	0.02	0.04	−0.05	−0.01	−0.07	−0.05	0.04
Italy	0.60	0.30	0.18	0.05	−0.14	−0.30	−0.39	−0.43	−0.41	−0.33	−0.21	−0.04

- The cross-correlations of output are high
- The cross-correlations of output are higher than the one of productivity
- The cross-correlations of productivity are higher than the cross-correlations of consumption.
- The cross-correlations of output, investment and employment are generally positive.
- See AMBLER, CARDIA and ZIMMERMANN, "International business cycles : What are the facts ?", Journal of Monetary Economics, 2004.

5. International Business Cycles

Between countries

Average cross-correlations

Variable	Averages from 190 cross-correlations		
	Full sample 1960:1–2000:4	1973:1–2000:4	1973:1–1990:4
Output	0.22	0.28	0.30
	(0.03)	(0.03)	(0.03)
	0.00	*0.00*	*0.00*
Consumption	0.14	0.15	0.14
	(0.02)	(0.03)	(0.03)
	0.00	*0.00*	*0.00*
Investment	0.18	0.22	0.22
	(0.04)	(0.04)	(0.03)
	0.00	*0.00*	*0.00*
Employment	0.20	0.22	0.21
	(0.03)	(0.03)	(0.04)
	0.00	*0.00*	*0.00*
Total hours	0.26	0.29	0.26
	(0.04)	(0.04)	(0.03)
	0.00	*0.00*	*0.00*
Employment[a]	0.25	0.26	0.25
	(0.04)	(0.04)	(0.05)
	0.00	*0.00*	*0.00*
Productivity	0.16	0.21	0.24
(from y and n only)	(0.02)	(0.02)	(0.03)
	0.00	*0.00*	*0.00*
Productivity	0.09	0.11	0.13
(best available)[b]	(0.02)	(0.02)	(0.02)
	0.00	*0.00*	*0.00*